

Research Interests

I work on *image/video processing*, *computer vision*, and *computational photography*, particularly on essential problems requiring *machine learning* with insights from *geometry* and *domain-specific knowledge*.

- Image and video synthesis and generation
- 3D from multi-view and sensors
- Low-level vision

Work Experience

- Feb. 2026 – **Department of Computer Science, National Yang Ming Chiao Tung University.**
Associate Professor
- Feb. 2023 – **Department of Computer Science, National Yang Ming Chiao Tung University.**
Feb. 2026 *Assistant Professor*
- May 2022 – **Meta Reality Labs Research.**
Oct. 2022 *Research Scientist Intern*
- Jun. 2017 – **MediaTek Inc.**
Feb. 2022 *Senior Algorithm Development Engineer*
- Sep. 2014 – **MediaTek Inc.**
Jun. 2017 *Algorithm Development Engineer*

Selected Honors and Awards

- Mar. 2025 **K. T. Li Young Researcher Award**, *Institute of Information & Computing Machinery (IICM).*
- Dec. 2024 **Outstanding Contribution Award for Young CS Scholars**, *College of CS, NYCU.*
- June 2024 **NSTC 2030 Emerging Young Scholar**, *National Science and Technology Council, R.O.C. (Taiwan).*
- May 2024 **CVPR 2024 Outstanding Reviewer.**
- April 2024 **Google Research Scholar Award.**
- Aug. 2023 **Google's Research Grant**, *Google University Relations Programs.*
- Aug. 2023 **Best Ph.D. Thesis Award**, *The Chinese Image Processing and Pattern Recognition Society (IPPR).*
- Jul. 2023 **Yushan Young Fellow Program**, *Ministry of Education, R.O.C. (Taiwan).*
- Nov. 2013 **Academic Achievement Award**, *Institute of Electronics, NCTU.*
- May. 2013 **Academic Achievement Award**, *Institute of Electronics, NCTU.*

Student Honors

- July 2026 **College Student Research Creativity Award**, *National Science and Technology Council (NSTC).*
- June 2026 **Hon Hai Technology Award**, *Hon Hai Education Foundation.*
- Jan. 2026 **3rd place of the NYCU CS Undergraduate Research Competition**, *CS, NYCU.*
- Dec. 2025 **Best Master Thesis Award**, *Taiwanese Association for Artificial Intelligence (TAAI).*
- Dec. 2025 **Excellent Master Thesis Award**, *Taiwanese Association for Artificial Intelligence (TAAI).*
- July 2025 **3rd place of the NYCU CS Undergraduate Research Competition**, *CS, NYCU.*
- July 2025 **2nd place of the NYCU CS Undergraduate Research Competition**, *CS, NYCU.*
- Nov. 2024 **Excellent Master Thesis Award**, *Taiwanese Association for Artificial Intelligence (TAAI).*
- Aug. 2024 **Excellent Master Thesis Award**, *The Chinese Image Processing and Pattern Recognition Society (IPPR).*
- July 2024 **3rd place of the NYCU CS Undergraduate Research Competition**, *CS, NYCU.*

July 2024 **2nd place of the NYCU CS Undergraduate Research Competition**, CS, NYCU.
Jan. 2024 **1st place of the NYCU CS Undergraduate Research Competition**, CS, NYCU.

Education

- 2017–2022 **Ph.D.**, *National Taiwan University*, Taipei, Taiwan.
Department of Computer Science and Information Engineering, CMLAB
Thesis: Restoring and Enhancing Images and Videos by Combining Modeling and Learning
Advisor: Yung-Yu Chuang [link](#)
- 2012–2014 **M.S.**, *National Chiao-Tung University*, Hsinchu, Taiwan.
Institute of Electronics, CommLab
Thesis: Wide-Angle Virtual View Synthesis with Depth-Based Background Modeling
Advisor: Hsueh-Ming Hang [link](#)
- 2008–2012 **B.S.**, *National Chiao-Tung University*, Hsinchu, Taiwan.
Department of Electronics Engineering

Publications [\(Google Scholar profile\)](#)

Journal papers

- IJCV 2024 **Portrait Distortion Correction with Perspective-Aware 3D GANs.**
Zhixiang Wang, [Yu-Lun Liu](#), Jia-Bin Huang, Shin'ichi Satoh, Sizhuo Ma, Guru Krishnan, Jian Wang
arXiv, 2023 [paper](#) [project](#)
- TPAMI 2021 **Learning to See Through Obstructions with Layered Decomposition.**
[Yu-Lun Liu](#), Wei-Sheng Lai, Ming-Hsuan Yang, Yung-Yu Chuang, and Jia-Bin Huang
IEEE Transactions on Pattern Analysis and Machine Intelligence, 2021 [paper](#) [project](#)

Conference papers

- ECCV 2026 **Scal3R: Learning Efficient Multi-Relative Pose Query for Scalable Online 3D Reconstruction.**
Chin-Yang Lin, Yang-Che Sun, Cheng Sun, Fu-En Yang, Min-Hung Chen, Yen-Yu Lin, Wei-Chen Chiu, [Yu-Lun Liu](#)
Proceedings of European Conference on Computer Vision, 2026
- ECCV 2026 **PriSM: Parsing and Style-Mixed Consistency for Unsupervised Domain Adaptation in Facial Landmark Detection.**
Chieh-Yu Yang, Hou-Ning Hu, Sykai Chen, [Yu-Lun Liu](#), Yen-Yu Lin
Proceedings of European Conference on Computer Vision, 2026
- ECCV 2026 **JanusMesh: Fast and Zero-Shot 3D Visual Illusion Generation via Cross-Space Denoising.**
Siang-Ling Zhang, Huai-Hsun Cheng, Tsung-Ju Yang, [Yu-Lun Liu](#)
Proceedings of European Conference on Computer Vision, 2026 [paper](#) [project](#)
- ECCV 2026 **DF3DV-1K: A Large-Scale Dataset and Benchmark for Distractor-Free Novel View Synthesis.**
Cheng-You Lu, Yi-Shan Hung, Wei-Ling Chi, Hao-Ping Wang, Charlie Li-Ting Tsai, Yu-Cheng Chang, [Yu-Lun Liu](#), Thomas Do, Chin-Teng Lin
Proceedings of European Conference on Computer Vision, 2026 [paper](#) [project](#)
- ECCV 2026 **RL-AWB: Deep Reinforcement Learning for Auto White Balance Correction in Low-Light Night-time Scenes.**
Yuan-Kang Lee, Kuan-Lin Chen, Chia-Che Chang, [Yu-Lun Liu](#)
Proceedings of European Conference on Computer Vision, 2026 [paper](#) [project](#)
- ECCV 2026 **Stream-DiffVSR: Low-Latency Streamable Video Super-Resolution via Auto-Regressive Diffusion.**
Hau-Shiang Shiu, Chin-Yang Lin, Zhixiang Wang, Chi-Wei Hsiao, Po-Fan Yu, Yu-Chih Chen, [Yu-Lun Liu](#)
Proceedings of European Conference on Computer Vision, 2026 [paper](#) [project](#)
- ECCV 2026 **Skyfall-GS: Synthesizing Immersive 3D Urban Scenes from Satellite Imagery.**
Jie-Ying Lee, Yi-Ruei Liu, Shr-Ruei Tsai, Wei-Cheng Chang, Chung-Ho Wu, Jiewen Chan, Zhenjun Zhao, Chieh Hubert Lin, [Yu-Lun Liu](#)
Proceedings of European Conference on Computer Vision, 2026 [paper](#) [project](#)
- SIGGRAPH 2026 **LongE2V: Long-Horizon Event-based Video Reconstruction, Prediction, and Frame Interpolation with Video Diffusion Models.**
Cheng-De Fan, Chun-Wei Tuan Mu, Chen-Wei Chang, Chin-Yang Lin, Kun-Ru Wu, Yu-Chee Tseng, [Yu-Lun Liu](#)
ACM SIGGRAPH 2026 Conference Papers, 2026 [paper](#) [project](#)

- SIGGRAPH 2026 **Stroke of Surprise: Progressive Semantic Illusions in Vector Sketching.**
Huai-Hsun Cheng, Siang-Ling Zhang, [Yu-Lun Liu](#)
ACM SIGGRAPH 2026 Conference Papers, 2026 [i](#) [paper](#) [i](#) [project](#)
- SIGGRAPH 2026 **VecSet-Edit: Unleashing Pre-trained LRM for Mesh Editing from Single Image.**
Teng-Fang Hsiao, Bo-Kai Ruan, [Yu-Lun Liu](#), Hong-Han Shuai
ACM SIGGRAPH 2026 Conference Papers, 2026 [i](#) [paper](#) [i](#) [project](#)
- CVPR Findings 2026 **AdaGaR: Adaptive Gabor Representation for Dynamic Scene Reconstruction.**
Jiewen Chan, Zhenjun Zhao, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 [i](#) [paper](#) [i](#) [project](#)
- CVPR Findings 2026 **Gen-n-Val: Agentic Image Data Generation and Validation.**
Jing-En Huang, I-Sheng Fang, Tzuhsuan Huang, [Yu-Lun Liu](#), Chih-Yu Wang, Jun-Cheng Chen
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2026 **Pantheon360: Taming Digital Twin Generation via 3D-Aware 360° Video Diffusion.**
Ting-Hsuan Chen*, Ying-Huan Chen*, Tao Tu, Jie-Ying Lee, Cho-Ying Wu, Fangzhou Lin, Hengyuan Zhang, David Paz, Xinyu Huang, Yuliang Guo†, [Yu-Lun Liu](#)†, Yue Wang†, Liu Ren
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2026 **Reflection Separation from a Single Image via Joint Latent Diffusion.**
Zheng-Hui Huang, Zhixiang Wang, [Yu-Lun Liu](#), Yung-Yu Chuang
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2026 **PhaSR: Generalized Image Shadow Removal with Physically Aligned Priors.**
Chia-Ming Lee, Yu-Fan Lin, Yu-Jou Hsiao, Jin-Hui Jiang, [Yu-Lun Liu](#), Chih-Chung Hsu
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2026 **ReflexSplit: Single Image Reflection Separation via Layer Fusion-Separation.**
Chia-Ming Lee, Yu-Fan Lin, Jin-Hui Jiang, Yu-Jou Hsiao, Chih-Chung Hsu, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2026 **Voxify3D: Pixel Art Meets Volumetric Rendering.**
Yi-Chuan Huang, Jiewen Chan, Hao-Jen Chien, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 [i](#) [paper](#) [i](#) [project](#)
- WACV 2026 **PS3: Part level instance segmentation in 3D.**
Hong-Xuan Yen, Chiamin Chen, Yanqing Wang, [Yu-Lun Liu](#), Min Sun
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2026 [i](#) [paper](#)
- WACV 2026 **HDR Reconstruction Boosting with Training-Free and Exposure-Consistent Diffusion.**
Yo-Tin Lin, Sykai Chen, Hou-Ning Hu, Yen-Yu Lin, [Yu-Lun Liu](#)
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2026 [i](#) [paper](#) [i](#) [project](#)
- WACV 2026 **TED-4DGS: Temporally Activated and Embedding-based Deformation for 4DGS Compression.**
Cheng-Yuan Ho, He-Bi Yang, Jui-Chiu Chiang, [Yu-Lun Liu](#), Wen-Hsiao Peng
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2026 [i](#) [paper](#) [i](#) [project](#)
- WACV 2026 **Splannequin: Freezing Monocular Mannequin-Challenge Footage with Dual-Detection Splatting.**
Hao-Jen Chien, Yi-Chuan Huang, Chung-Ho Wu, Wei-Lun Chao, [Yu-Lun Liu](#)
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2026 [i](#) [paper](#) [i](#) [project](#)
- SIGGRAPH Asia 2025 **Prior-Enhanced Gaussian Splatting for Dynamic Scene Reconstruction from Casual Video.**
Meng-Li Shih, Ying-Huan Chen, [Yu-Lun Liu](#), Brian Curless,
ACM SIGGRAPH Asia 2025 Conference Papers, 2025 [i](#) [paper](#) [i](#) [project](#)
- NeurIPS 2025 **FIPER: Generalizable Factorized Fields for Joint Image Compression and Super-Resolution.**
Yang-Che Sun, Cheng Yu Yeo, Ernie Chu, Jun-Cheng Chen, [Yu-Lun Liu](#)
Conference on Neural Information Processing Systems, 2025 [i](#) [paper](#) [i](#) [project](#)
- ICCV 2025 **LightsOut: Diffusion-based Outpainting for Enhanced Lens Flare Removal.**
Shr-Ruei Tsai, Wei-Cheng Chang, Jie-Ying Lee, Chih-Hai Su, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision, 2025 [i](#) [paper](#) [i](#) [project](#)
- ICCV 2025 **StealthAttack: Robust 3D Gaussian Splatting Poisoning via Density-Guided Illusions.**
Bo-Hsu Ke, You-Zhe Xie, [Yu-Lun Liu](#), Wei-Chen Chiu
Proceedings of the IEEE/CVF Conference on Computer Vision, 2025 [i](#) [paper](#) [i](#) [project](#)
- ICCV 2025 **LongSplat: Robust Unposed 3D Gaussian Splatting for Casual Long Videos.**
Chin-Yang Lin, Cheng Sun, Fu-En Yang, Min-Hung Chen, Yen-Yu Lin, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision, 2025 [i](#) [paper](#) [i](#) [project](#)

- ICCV 2025 **OpenM3D: Open Vocabulary Multi-view Indoor 3D Object Detection without Human Annotations.**
Peng-Hao Hsu, Ke Zhang, Fu-En Wang, Tao Tu, Ming-Feng Li, [Yu-Lun Liu](#), Albert Y. C. Chen, Min Sun, Cheng-Hao Kuo
Proceedings of the IEEE/CVF Conference on Computer Vision, 2025 [i](#) [paper](#)
- CoRL 2025 **See, Point, Fly: A Learning-Free VLM Framework for Universal Unmanned Aerial Navigation.**
Chih Yao Hu, Yang-Sen Lin, Yuna Lee, Chih-Hai Su, Jie-Ying Lee, Shr-Ruei Tsai, Chin-Yang Lin, Kuan-Wen Chen, Tsung-Wei Ke, [Yu-Lun Liu](#)
9th Annual Conference on Robot Learning, 2025 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2025 **GCC: Generative Color Constancy via Diffusing a Color Checker.**
Chen-Wei Chang, Cheng-De Fan, Chia-Che Chang, Yi-Chen Lo, Yu-Chee Tseng, Jiun-Long Huang, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2025 **AuraFusion360: Augmented Unseen Region Alignment for Reference-based 360° Unbounded Scene Inpainting.**
Chung-Ho Wu*, Yang-Jung Chen*, Ying-Huan Chen, Jie-Ying Lee, Bo-Hsu Ke, Chun-Wei Tuan Mu, Yi-Chuan Huang, Chin-Yang Lin, Min-Hung Chen, Yen-Yu Lin, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2025 **SpectroMotion: Dynamic 3D Reconstruction of Specular Scenes.**
Cheng-De Fan, Chen-Wei Chang, Yi-Ruei Liu, Jie-Ying Lee, Jiun-Long Huang, Yu-Chee Tseng, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2025 **FrugalNeRF: Fast Convergence for Few-shot Novel View Synthesis without Learned Priors.**
Chin-Yang Lin*, Chung-Ho Wu*, Chang-Han Yeh, Shih-Han Yen, Cheng Sun, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2025 **DeNVer: Deformable Neural Vessel Representations for Unsupervised Video Vessel Segmentation.**
Chun-Hung Wu, Shih-Hong Chen, Chih-Yao Hu, Hsin-Yu Wu, Kai-Hsin Chen, Yu-You Chen, Chih-Hai Su, Chih-Kuo Lee, [Yu-Lun Liu](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025 [i](#) [paper](#) [i](#) [project](#)
- ICLR 2025 **CAT-3DGS: A Context-Adaptive Triplane Approach to Rate-Distortion-Optimized 3DGS Compression.**
Yu-Ting Zhan*, Cheng-Yuan Ho*, Hebi Yang, Jui-Chiu Chiang, [Yu-Lun Liu](#), Wen-Hsiao Peng
International Conference on Learning Representations, 2025 [i](#) [paper](#)
- ICASSP 2025 **3D Gaussian Splatting with Grouped Uncertainty for Unconstrained Images.**
Hao-Yu Hou*, Chia-Chi Hsu*, Yu-Chen Huang*, Mu-Yi Shen*, Wei-Fang Sun, Cheng Sun, Chia-Che Chang, [Yu-Lun Liu](#), Chun-Yi Lee
IEEE International Conference on Acoustics, Speech, and Signal Processing, 2025 [i](#) [paper](#)
- WACV 2025 **ORFormer: Occlusion-Robust Transformer for Accurate Facial Landmark Detection.**
Jui-Che Chiang, Hou-Ning Hu, Bo-Syuan Hou, Chia-Yu Tseng, [Yu-Lun Liu](#), Min-Hung Chen, Yen-Yu Lin
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2025 [i](#) [paper](#) [i](#) [project](#)
[Oral Presentation](#)
- WACV 2025 **CorrFill: Enhancing Faithfulness in Reference-based Inpainting with Correspondence Guidance in Diffusion Models.**
Kuan-Hung Liu, Cheng-Kun Yang, Min-Hung Chen, [Yu-Lun Liu](#), Yen-Yu Lin
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2025 [i](#) [paper](#) [i](#) [project](#)
[Oral Presentation](#)
- NeurIPS 2024 **ReF-LDM: A Latent Diffusion Model for Reference-based Face Image Restoration.**
Chi-Wei Hsiao, [Yu-Lun Liu](#), Cheng-Kun Yang, Sheng-Po Kuo, Yucheun Kevin Jou, Chia-Ping Chen
Conference on Neural Information Processing Systems, 2024 [i](#) [paper](#) [i](#) [project](#)
- NeurIPS 2024 **Depth Anywhere: Enhancing 360 Monocular Depth Estimation via Perspective Distillation and Unlabeled Data Augmentation.**
Ning-Hsu Wang, [Yu-Lun Liu](#)
Conference on Neural Information Processing Systems, 2024 [i](#) [paper](#) [i](#) [project](#)
- NeurIPS 2024 **NaRCan: Natural Refined Canonical Image with Integration of Diffusion Prior for Video Editing.**
Ting-Hsuan Chen, Jiewen Chan, Hau-Shiang Shiu, Shih-Han Yen, Chang-Han Yeh, [Yu-Lun Liu](#)
Conference on Neural Information Processing Systems, 2024 [i](#) [paper](#) [i](#) [project](#)
- IROS 2024 **Precise Pick-and-Place using Score-Based Diffusion Networks.**
Shih-Wei Guo, Tsu-Ching Hsiao, [Yu-Lun Liu](#), Chun-Yi Lee
IEEE/RSJ International Conference on Intelligent Robots and Systems, 2024 [i](#) [paper](#) [i](#) [project](#)

- ECCV 2024 **GenRC: 3D Indoor Scene Generation from Sparse Image Collections.**
Ming-Feng Li, Yueh-Feng Ku, Hong-Xuan Yen, Chi Liu, [Yu-Lun Liu](#), Albert Y. C. Chen, Cheng-Hao Kuo, Min Sun
Proceedings of European Conference on Computer Vision, 2024 [i](#) [paper](#) [i](#) [project](#)
- ICRA Workshop 2024 **DriveEnv-NeRF: Exploration of A NeRF-Based Autonomous Driving Environment for Real-World Performance Validation.**
Mu-Yi Shen, Chia-Chi Hsu, Hao-Yu Hou, Yu-Chen Huang, Wei-Fang Sun, Chia-Che Chang, [Yu-Lun Liu](#), Chun-Yi Lee
IEEE International Conference on Robotics and Automation RoboNeRF Workshop, 2024 [i](#) [paper](#) [i](#) [project](#)
- SIGGRAPH 2024 **Matting by Generation.**
Zhixiang Wang, Baiang Li, Jian Wang, [Yu-Lun Liu](#), Jinwei Gu, Yung-Yu Chuang, Shin'ichi Satoh
ACM SIGGRAPH 2024 Conference Papers, 2024 [i](#) [paper](#) [i](#) [project](#)
- SIGGRAPH 2024 **BoostMVSNeRFs: Boosting MVS-based NeRFs to Generalizable View Synthesis in Large-scale Scenes.**
Chih-Hai Su*, Chih-Yao Hu*, Shr-Ruei Tsai*, Jie-Ying Lee*, Chin-Yang Lin, [Yu-Lun Liu](#)
ACM SIGGRAPH 2024 Conference Papers, 2024 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2024 **Image-Text Co-Decomposition for Text-Supervised Semantic Segmentation.**
Ji-Jia Wu, Andy Chia-Hao Chang, Chieh-Yu Chuang, Chun-Pei Chen, [Yu-Lun Liu](#), Min-Hung Chen, Hou-Ning Hu, Yung-Yu Chuang, Yen-Yu Lin
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024 [i](#) [paper](#)
- CVPR 2024 **HumanNeRF-SE: A Simple yet Effective Approach to Animate HumanNeRF with Diverse Poses.**
Caoyuan Ma, [Yu-Lun Liu](#), Zhixiang Wang, Wu Liu, Xinchun Liu, Zheng Wang
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024 [i](#) [paper](#) [i](#) [project](#)
- ICLR 2024 **Dual Associated Encoder for Face Restoration.**
Yu-Ju Tsai, [Yu-Lun Liu](#), Lu Qi, Kelvin C.K. Chan, Ming-Hsuan Yang
The Twelfth International Conference on Learning Representations, 2024 [i](#) [paper](#) [i](#) [project](#)
- AAAI 2024 **Improving Robustness for Joint Optimization of Camera Poses and Decomposed Low-Rank Tensorial Radiance Fields.**
Bo-Yu Cheng, Wei-Chen Chiu, [Yu-Lun Liu](#)
Proceedings of AAAI Conference on Artificial Intelligence, 2024 [i](#) [paper](#) [i](#) [project](#)
- ICCV 2023 **Learning Continuous Exposure Value Representations for Single-Image HDR Reconstruction.**
Su-Kai Chen, Hung-Lin Yen, [Yu-Lun Liu](#), Min-Hung Chen, Hou-Ning Hu, Wen-Hsiao Peng and Yen-Yu Lin
Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023 [i](#) [project](#)
- ICCV 2023 **ImGeoNet: Image-induced Geometry-aware Voxel Representation for Multi-view 3D Object Detection.**
Tao Tu, Shun-Po Chuang, [Yu-Lun Liu](#), Cheng Sun, Ke Zhang, Donna Roy, Cheng-Hao Kuo, Min Sun
Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023 [i](#) [project](#)
- CVPR 2023 **Progressively Optimized Local Radiance Fields for Robust View Synthesis.**
Andreas Meuleman, [Yu-Lun Liu](#), Chen Gao, Jia-Bin Huang, Changil Kim, Min H. Kim, Johannes Kopf
Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2023 **Robust Dynamic Radiance Fields.**
[Yu-Lun Liu](#), Chen Gao, Andreas Meuleman, Hung-Yu Tseng, Ayush Saraf, Changil Kim, Yung-Yu Chuang, Johannes Kopf, Jia-Bin Huang
Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023 [i](#) [paper](#) [i](#) [project](#)
- ICLR 2022 **Denoising Likelihood Score Matching for Conditional Score-based Data Generation .**
Chen-Hao Chao, Wei-Fang Sun, Bo-Wun Cheng, Yi-Chen Lo, Chia-Che Chang, [Yu-Lun Liu](#), Yu-Lin Chang, Chia-Ping Chen, and Chun-Yi Lee
International Conference on Learning Representations, 2022 [i](#) [paper](#)
- ICCV 2021 **Bridging Unsupervised and Supervised Depth from Focusvia All-in-Focus Supervision.**
Ning-Hsu Wang, Ren Wang, [Yu-Lun Liu](#), Yu-Hao Huang, Yu-Lin Chang, Chia-Ping Chen, and Kevin Jou
Proceedings of IEEE/CVF Conference on Computer Vision, 2021 [i](#) [paper](#) [i](#) [project](#)
- ICCV 2021 **Hybrid Neural Fusion for Full-frame Video Stabilization.**
[Yu-Lun Liu](#), Wei-Sheng Lai, Ming-Hsuan Yang, Yung-Yu Chuang, and Jia-Bin Huang
Proceedings of IEEE/CVF Conference on Computer Vision, 2021 [i](#) [paper](#) [i](#) [project](#)
- ICPR 2020 **Explorable Tone Mapping Operators.**
Chien-Chuan Su, Ren Wang, Hung-Jin Lin, [Yu-Lun Liu](#), Chia-Ping Chen, Yu-Lin Chang, and Soo-Chang Pei
Proceedings of the 25th International Conference on Pattern Recognition, 2020 [i](#) [paper](#)

- ECCV 2020 **Learning Camera-Aware Noise Models.**
Ke-Chi Chang, Ren Wang, Hung-Jin Lin, [Yu-Lun Liu](#), Chia-Ping Chen, Yu-Lin Chang, and Hwann-Tzong Chen
Proceedings of European Conference on Computer Vision, 2020 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2020 **Learning to See Through Obstructions.**
[Yu-Lun Liu](#), Wei-Sheng Lai, Ming-Hsuan Yang, Yung-Yu Chuang, and Jia-Bin Huang
Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020 [i](#) [paper](#) [i](#) [project](#)
- CVPR 2020 **Single-Image HDR Reconstruction by Learning to Reverse the Camera Pipeline.**
[Yu-Lun Liu](#)*, Wei-Sheng Lai*, Yu-Sheng Chen, Yi-Lung Kao, Ming-Hsuan Yang, Yung-Yu Chuang, and Jia-Bin Huang
Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020 [i](#) [paper](#) [i](#) [project](#)
- AAAI 2020 **Attention-based View Selection Networks for Light-field Disparity Estimation.**
Yu-Ju Tsai, [Yu-Lun Liu](#), Yung-Yu Chuang, and Ming Ouhyoung
Proceedings of AAAI Conference on Artificial Intelligence, 2020 [i](#) [paper](#)
- AAAI 2019 **Deep Video Frame Interpolation using Cyclic Frame Generation.**
[Yu-Lun Liu](#), Yi-Tung Liao, Yen-Yu Lin, and Yung-Yu Chuang
Proceedings of AAAI Conference on Artificial Intelligence, 2019 [i](#) [paper](#) [i](#) [project](#)
Oral Presentation
- APSIPA 2014 **Background modeling using depth information.**
[Yu-Lun Liu](#) and Hsueh-Ming Hang
Signal and Information Processing Association Annual Summit and Conference, 2014
- PCS 2013 **Virtual view synthesis using backward depth warping algorithm.**
Du-Hsiu Li, Hsueh-Ming Hang, and [Yu-Lun Liu](#)
Picture Coding Symposium, 2013
- [Preprints](#)
- arXiv 2024 **DiffIR2VR-Zero: Zero-Shot Video Restoration with Diffusion-based Image Restoration Models.**
Chang-Han Yeh, Chin-Yang Lin, Zhixiang Wang, Chi-Wei Hsiao, Ting-Hsuan Chen, [Yu-Lun Liu](#)
arXiv, 2024 [i](#) [paper](#) [i](#) [project](#)

Talks

- Invited Talk **Generative AI for Immersive 3D Worlds.**
Beyond Reality: VR × Generative AI Workshop, NYCU, Hsinchu, Taiwan, June 2026.
- Invited Talk **From Casual Capture to Deployable 3D: Reconstruction & Understanding with 3D Foundation Models.**
NVIDIA GTC Taipei 2026 Research Summit, Taipei, Taiwan, June 2026.
- Invited Seminar **Breaking Through 3DGS: Tackling Dynamics, Editing, Long Videos, and Security.**
Department of Electrical Engineering, NTHU, Hsinchu, Taiwan, Nov. 2025.
- Guest Lecture **3D from Multi-view.**
MediaTek Inc., Hsinchu, Taiwan, July 2025.
- Invited Talk **What School and Work Didn't Teach Me.**
Library, NYCU, Hsinchu, Taiwan, Sep. 2024.
- Invited Seminar **Enhancing 3D Scene Reconstruction in NeRFs.**
CSIE, NTU, Taipei, Taiwan, Mar. 2024.
- Keynote **AI Photographer: How Machines Are Changing the Way We See the World.**
Fun AI Winter Camp, NCKU, Tainan, Taiwan, Feb. 2024.
- Invited Talk **Dynamic and Local Radiance Fields for Robust View Synthesis.**
MediaTek Inc., Hsinchu, Taiwan, May. 2023.
- Invited Talk **Dynamic and Local Radiance Fields for Robust View Synthesis.**
Institute of Communications Engineering, NTHU, Hsinchu, Taiwan, April. 2023.
- Invited Talk **Dynamic and Local Radiance Fields for Robust View Synthesis.**
The 5th Augmented Intelligence and Interaction (AII) Workshop, Taoyuan, Taiwan, March. 2023.
- Guest Lecture **What School and the Workplace Didn't Teach Me.**
NYCU, Hsinchu, Taiwan, March. 2023.

- Invited Talk **Learning to See the Unseen.**
Taichung Municipal Taichung First Senior High School, Taichung, Taiwan, March. 2023.
- Invited Talk **Learning to See the Unseen.**
Research Center for Information Technology Innovation, Academia Sinica, Taipei, Taiwan, January. 2023.
- Invited Talk **Learning to See the Unseen.**
Institute of Information Systems and Applications, NTHU, Hsinchu, Taiwan, November. 2022.
- Guest Lecture **Recent Progress on Video Restoration.**
CommE7005, NTU, Taipei, Taiwan, Mar. 2022.
- Guest Lecture **Rethinking Multimedia and IC Design in the Era of AI.**
CommE7005, NTU, Taipei, Taiwan, Nov. 2021.
- Guest Lecture **CV Final Project - Video Frame Interpolation.**
EEE5053, NTU, Taipei, Taiwan, May. 2021.
- Guest Lecture **Rethinking Multimedia and IC Design in the Era of AI.**
DEE4592, NCTU, Hsinchu, Taiwan, Mar. 2021.
- Invited Talk **Learning to See Through Obstructions.**
2020 CVPR Taiwan Sharing Session, NTU, Taipei, Taiwan, July. 2020.
- Invited Talk **Single-Image HDR Reconstruction by Learning to Reverse the Camera Pipeline.**
2020 CVPR Taiwan Sharing Session, NTU, Taipei, Taiwan, July. 2020.
- Guest Lecture **CV Final Project - Depth Estimation.**
EEE5053, NTU, Taipei, Taiwan, Jan. 2020.
- Invited Talk **Deep Video Frame Interpolation using Cyclic Frame Generation.**
The 3rd Augmented Intelligence and Interaction (AII) Workshop, Taoyuan, Taiwan, July. 2019.
- Oral **Deep Video Frame Interpolation using Cyclic Frame Generation.**
AAAI, Honolulu, Hawaii, USA, Jan. 2019.
- Guest Lecture **CV Final Project - Depth Estimation.**
EEE5053, NTU, Taipei, Taiwan, Jan. 2019.

Academic Services

- Publication Chair ACCV 2024
- COI Coordinator SIGGRAPH Asia 2025
- Area Chair NeurIPS 2025, 2026
MVA 2025
- International Program Committee Pacific Graphics 2025
- Organizer 2020 CVPR Taiwan Sharing Session
- Thesis Committees Hsuan-Kung Yang (advisor: Chun-Yi Lee) - PhD, expected 2024
Yi-Hsin Chen (advisor: Wen-Hsiao Peng) - PhD, expected 2024
Ardian Umam (advisors: Yen-Yu Lin and Jen-Hui Chuang) - PhD, Dec 2024
Cheng Sun (advisors: Hwann-Tzong Chen and Min Sun) - PhD, Jul 2023
Hitika Tiwari (advisors: K. S. Venkatesh and Yong-Sheng Chen) - PhD, Jul 2023
Cheng-Kun Yang (advisors: Yung-Yu Chuang and Yen Yu-Lin) - PhD, Jul 2023

Journal Reviewer	IEEE Transactions on Image Processing (TIP) ACM Computing Surveys (CSUR) Applied Soft Computing Pattern Recognition ACM Transactions on Graphics (TOG) Information Fusion IEEE Transactions on Multimedia International Journal of Computer Vision The Journal of Supercomputing Computer Vision and Image Understanding Computers & Graphics Multidimensional Systems and Signal Processing International Journal of Pattern Recognition and Artificial Intelligence
Conference Reviewer	IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022, 2023, 2024, 2025 IEEE Conference on Computer Vision (ICCV), 2021, 2023, 2025 European Conference on Computer Vision (ECCV), 2020, 2022, 2024 British Machine Vision Conference (BMVC), 2023, 2024 Asian Conference on Computer Vision (ACCV), 2020, 2022, 2024 Conference on Neural Information Processing Systems (NeurIPS), 2022, 2023, 2024 International Conference on Learning Representations (ICLR), 2022, 2023, 2024, 2025 International Conference on Machine Learning (ICML), 2023, 2024 ACM SIGGRAPH (SIGGRAPH), 2022, 2023, 2024, 2025 ACM SIGGRAPH Asia (SIGGRAPH), 2024, 2025 AAAI Conference on Artificial Intelligence (AAAI), 2021, 2022, 2023, 2024, 2025 International Joint Conference on Artificial Intelligence (IJCAI), 2021, 2022, 2023, 2024, 2025 ACM International Conference on Multimedia (MM), 2018 IEEE International Conference on Multimedia and Expo Workshops (ICMEW), 2023 International Conference on Robotics and Automation (ICRA), 2023 Conference on Robot Learning (CoRL), 2025
Research Mentor	Research Mentor for interns at MediaTek Inc., 2019, 2020, 2021, 2022.
Professional Societies	IEEE student member AAAI student member TAAI member IPPR member

Teaching Experiences

Spring 2026	CSCS10017 Signals and Systems , <i>Department of Computer Science, National Yang Ming Chiao Tung University.</i>
Fall 2025	CSIC30191 Image and Video Generations , <i>Department of Computer Science, National Yang Ming Chiao Tung University.</i>
Fall 2025	CSIC30107 Video Compression , <i>Department of Computer Science, National Yang Ming Chiao Tung University.</i>
Spring 2025	CSCS10017 Signals and Systems , <i>Department of Computer Science, National Yang Ming Chiao Tung University.</i>
Fall 2024	CSIC30107 Video Compression , <i>Department of Computer Science, National Yang Ming Chiao Tung University.</i>
Spring 2024	CSCS10017 Signals and Systems , <i>Department of Computer Science, National Yang Ming Chiao Tung University.</i>
Fall 2023	CSIC30107 Video Compression , <i>Department of Computer Science, National Yang Ming Chiao Tung University.</i>
Spring 2023	CSIC30107 Video Compression , <i>Department of Computer Science, National Yang Ming Chiao Tung University.</i>

Spring 2013 **Teaching Assistant of DEE1315 Probability and Statistics with Prof. Hsueh-Ming Hang**, *Department of Electronics Engineering, National Chiao Tung University.*

References

- Ph.D. Advisor **Yung-Yu Chuang**, *Professor of Department of Computer Science and Information Engineering, National Taiwan University.*
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